Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.25	20.08	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.01	-0.8%
power (W), mean	628	643	+2.4%
power (W), peak	650	688	+6.0%
energy (Wh)	212	215	+1.5%
energy, parked workers still on at idle power (Wh)	212	215	+1.5%
CPU used (cores), mean	0.872	1.27	+45.9%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.036	0.053	+45.9%
energy per core-hour (Wh)	721	506	-29.8%
node-hours per core-hour	6.88	4.72	-31.5%
pending pods, pod-minutes	0.533	0.000	-100.0%
pending pods, peak	2.00	0.000	-100.0%
HPA replicas, mean	8.93	9.50	+6.5%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.517	0.517	+0.0%
requests timed	1,040	1,095	+5.3%
time over the response line (% of samples)	0.962	0.000	-100.0%
response time (ms), mean	156	100	-35.9%
response time (ms), median	130	90.90	-29.9%
response time (ms), 95th percentile	314	143	-54.7%
response time (ms), 99th percentile	488	178	-63.6%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.5719	5.1734	-31.7%	[-4.0150, -0.7241]	GAIN
kwh_per_core_hour	0.7895	0.5511	-30.2%	[-0.3980, -0.0733]	GAIN
utilisation	0.0359	0.0538	+50.0%	[+0.0047, +0.0327]	GAIN
pending_pod_minutes_per_hour	1.5238	0.0000	-100.0%	[-4.5714, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.4706	1.6466	+12.0%	[-2.2325, +2.6783]	EVEN, no significant difference

What Omni-Compass did

• decisions: 80 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	622	5.00	6.00	0.000	0.000	0.542	0.710
1	6.00	6.00	620	625	5.00	8.00	0.000	0.000	0.623	0.789
2	6.00	6.00	616	624	5.00	8.00	0.000	0.000	0.542	0.754
3	6.00	6.00	622	636	5.25	8.50	0.500	0.000	0.692	1.07
4	6.00	6.00	636	648	9.00	10.00	0.000	0.000	1.10	1.40
5	6.00	6.00	634	645	10.00	10.00	0.000	0.000	1.04	1.34
6	6.00	6.00	634	647	10.00	10.00	0.000	0.000	1.04	1.39
7	6.00	6.00	647	670	10.00	10.00	0.000	0.000	1.36	1.99
8	6.00	6.00	644	679	10.00	10.00	0.000	0.000	1.29	2.23
9	6.00	6.00	649	678	10.00	10.00	0.000	0.000	1.41	2.22
10	6.00	6.00	639	670	10.00	10.00	0.000	0.000	1.15	1.97
11	6.00	6.00	618	624	10.00	10.00	0.000	0.000	0.588	0.735
12	6.00	6.00	618	626	10.00	10.00	0.000	0.000	0.577	0.808
13	6.00	6.00	624	634	10.00	10.00	0.000	0.000	0.739	1.04
14	6.00	6.00	634	649	10.00	10.00	0.000	0.000	1.02	1.42
15	6.00	6.00	630	650	10.00	10.00	0.000	0.000	0.940	1.44
16	6.00	6.00	632	648	10.00	10.00	0.000	0.000	0.963	1.42
17	6.00	6.00	625	640	10.00	10.00	0.000	0.000	0.771	1.16
18	6.00	6.00	620	622	10.00	10.00	0.000	0.000	0.637	0.705
19	6.00	6.00	619	627	10.00	10.00	0.000	0.000	0.605	0.851

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.